Project Documentation: Automated Network Request Management

ServiceNow Developer Portfolio | K Tanuja

1. Functional Overview

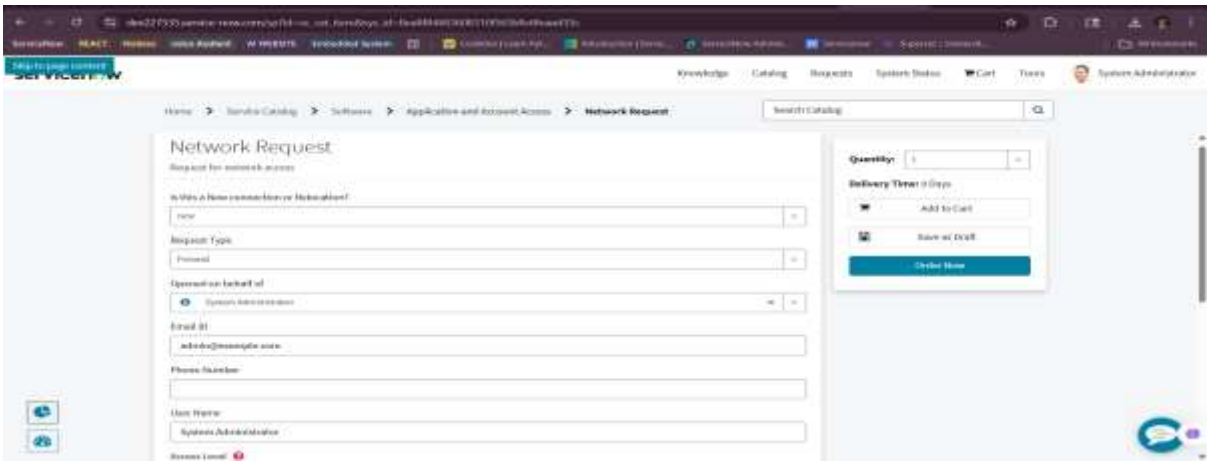
Goal: To modernize IT operations by automating manual network infrastructure requests.

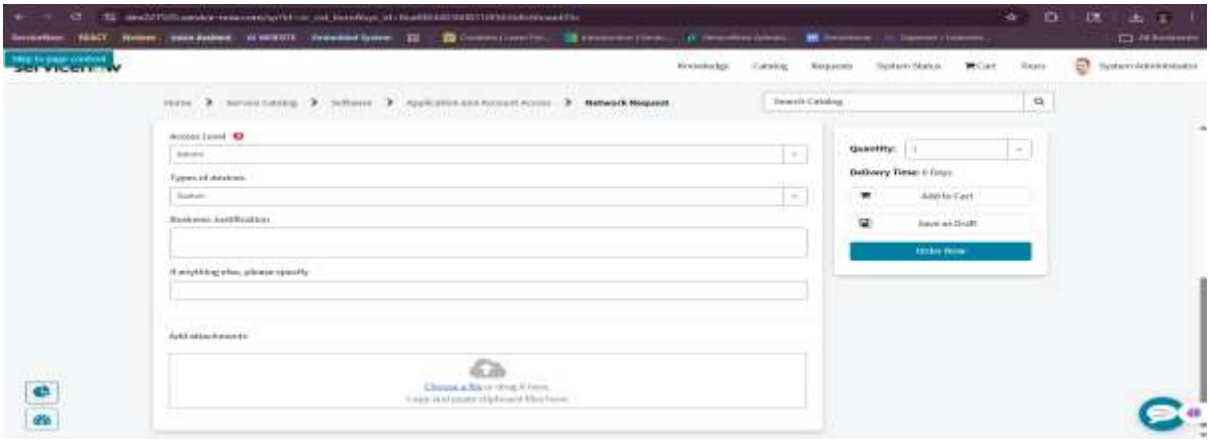
User Story & Experience

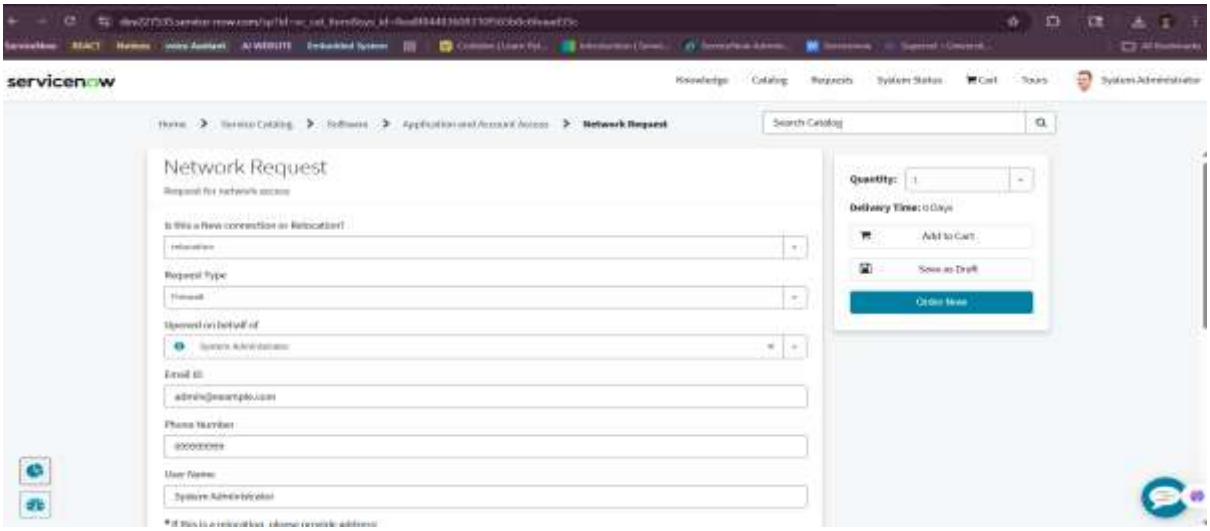
- **Self-Service:** Users submit requests via the Service Portal.
- **Intelligent Forms:** Mandatory fields and data validation prevent incomplete data.
- **Automated Routing:** Eliminates manual email chains for approvals.

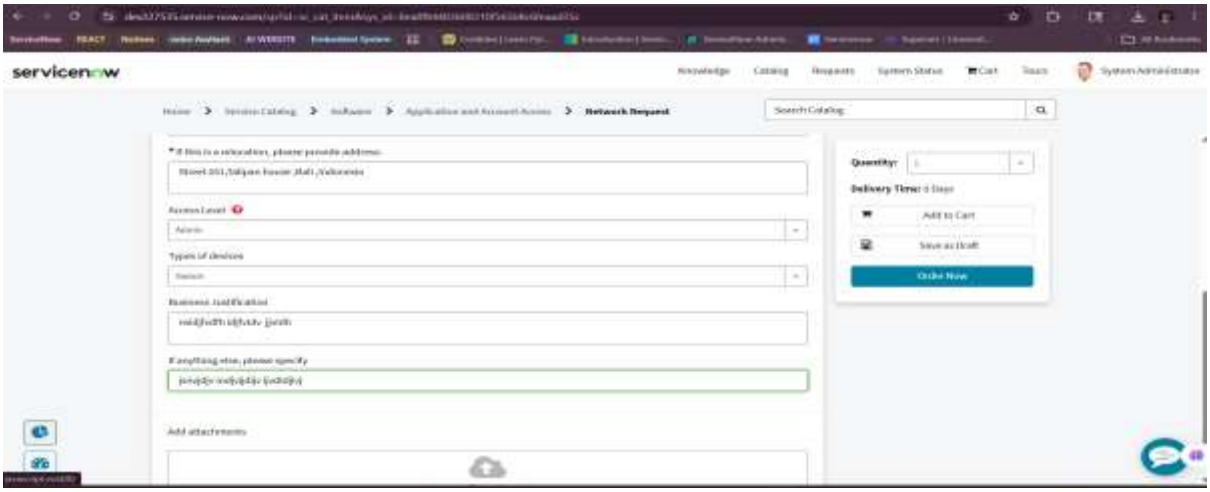
2.Technical Blueprint

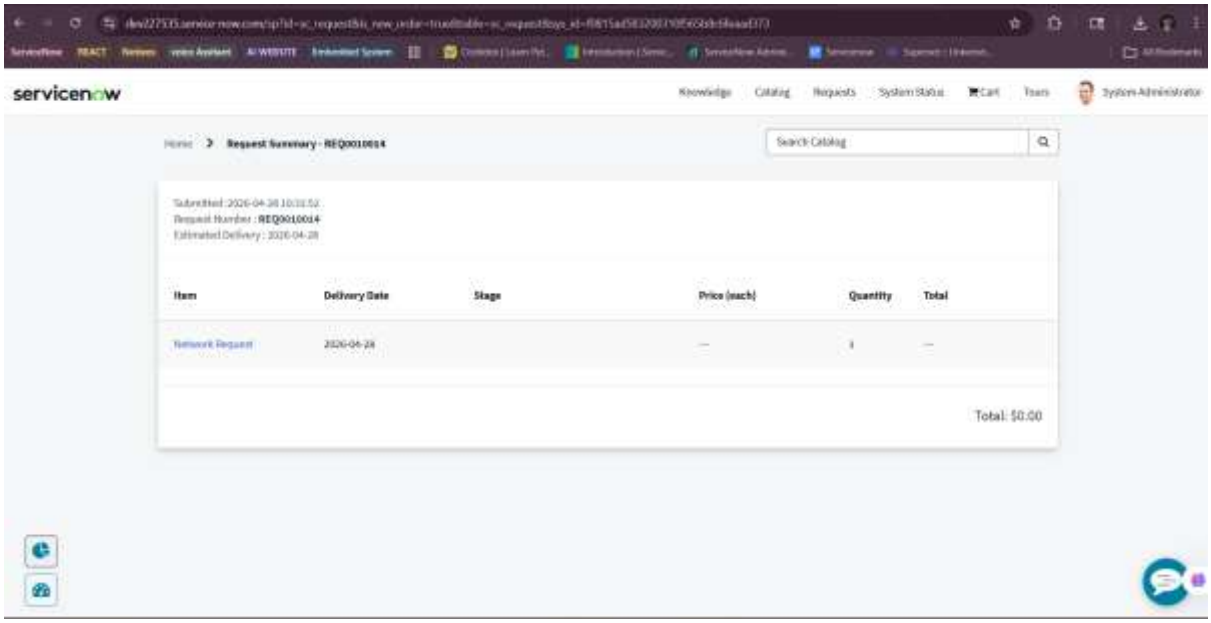
This section details the architecture used to map Service portal variables to back-end structured data.

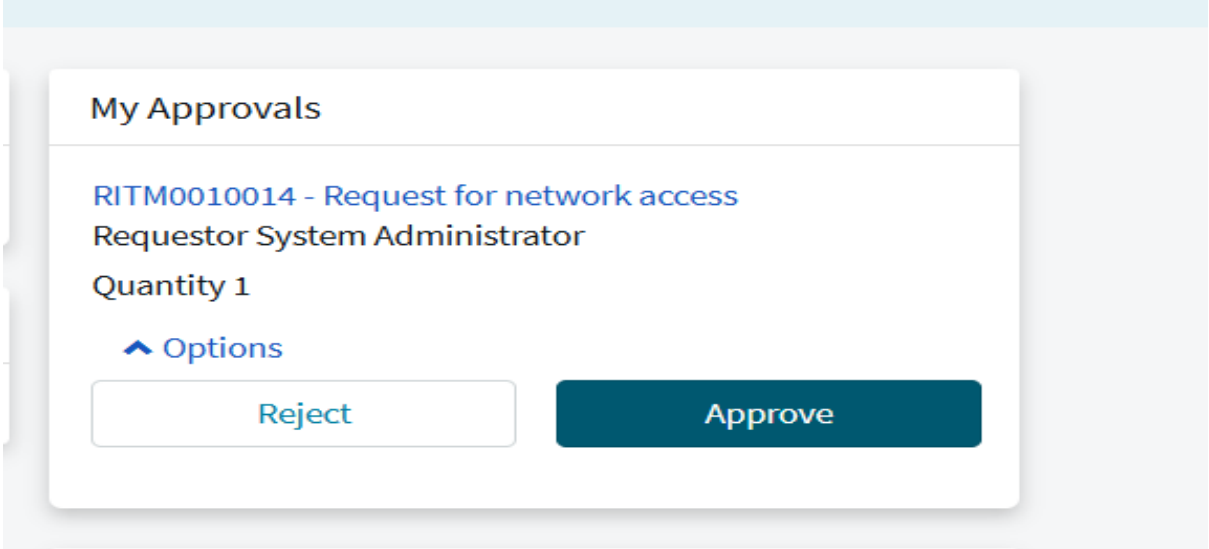


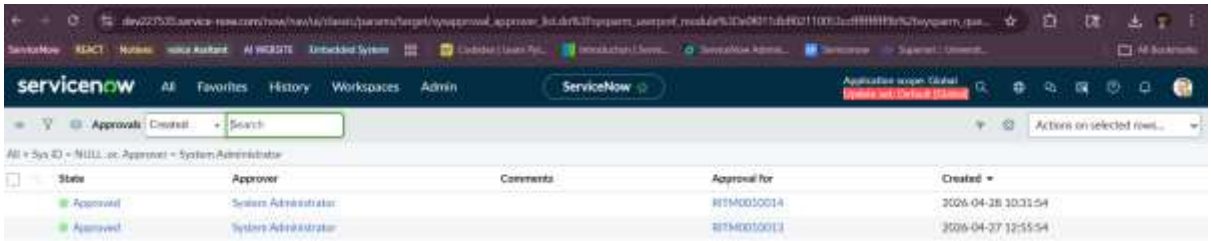

servicenow
Network Request
Quantity:
Delivery Time:
Add to Cart
Save as Draft
Order Now


servicenow
Network Request
Quantity:
Delivery Time:
Add to Cart
Save as Draft
Order Now


servicenow
Quantity:
Delivery Time:
Add to Cart
Save as Draft
Order Now

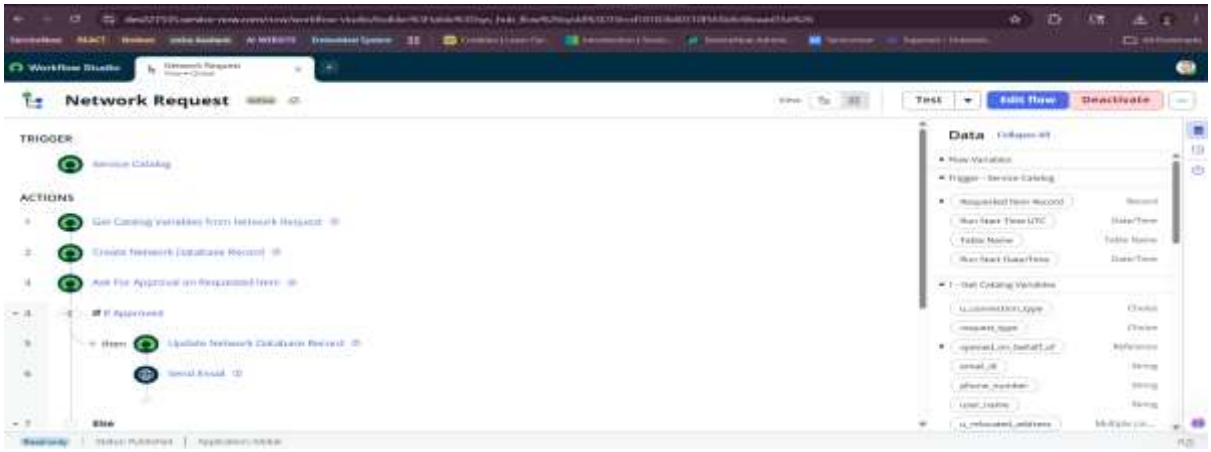

servicenow
Knowledge
Catalog
Requests
System Status
Cart
Tours
System Administrator
Home
Request Summary - REQ0010014
Search Catalog
Submitted : 2026-04-28 10:31:52
Request Number : REQ0010014
Estimated Delivery : 2026-04-28
Item
Delivery Date
Stage
Price (each)
Quantity
Total
Network Request
2026-04-28
Total: $0.00

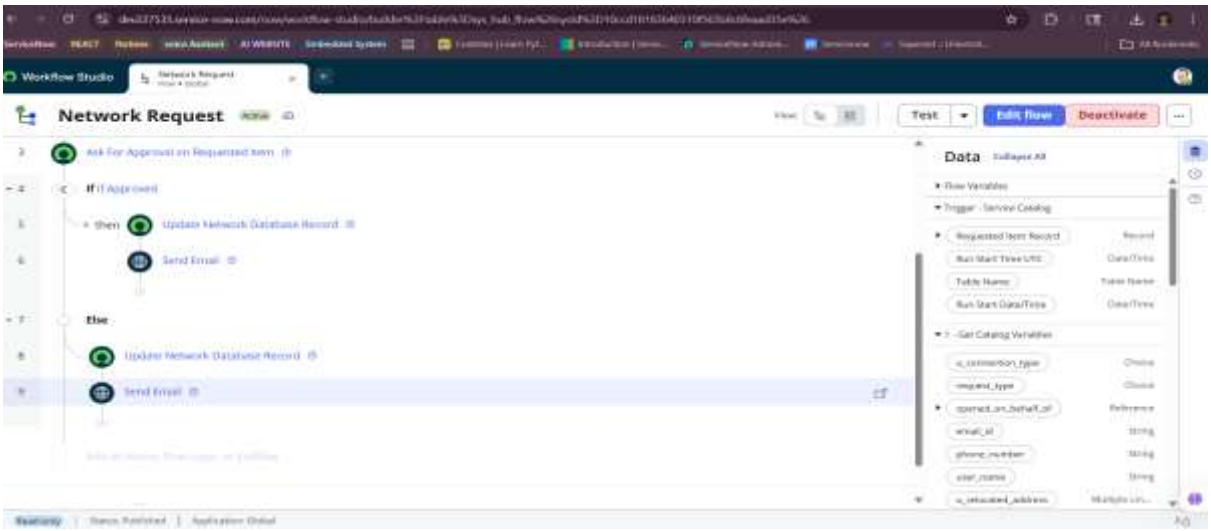

My Approvals
RITM0010014 - Request for network access
Requestor System Administrator
Quantity 1
Options
Reject
Approve

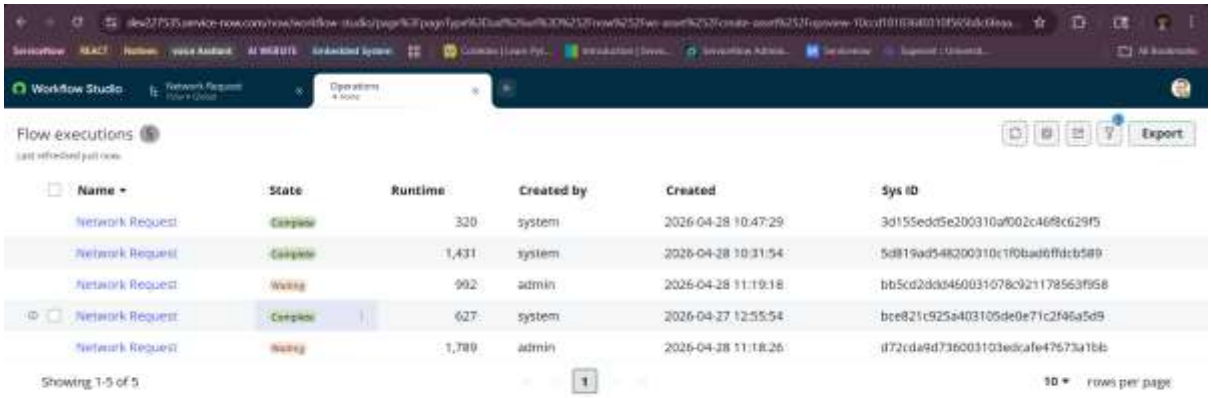

servicenow
All
Favorites
History
Workspaces
Admin
ServiceNow
Approvals
Search
Actions on selected rows...
All = Sys ID = NULL or Approver = System Administrator
State
Approver
Comments
Approval for
Created
Approved
System Administrator
RITM0010014
2026-04-28 10:31:54
Approved
System Administrator
RITM0010013
2026-04-27 12:55:54

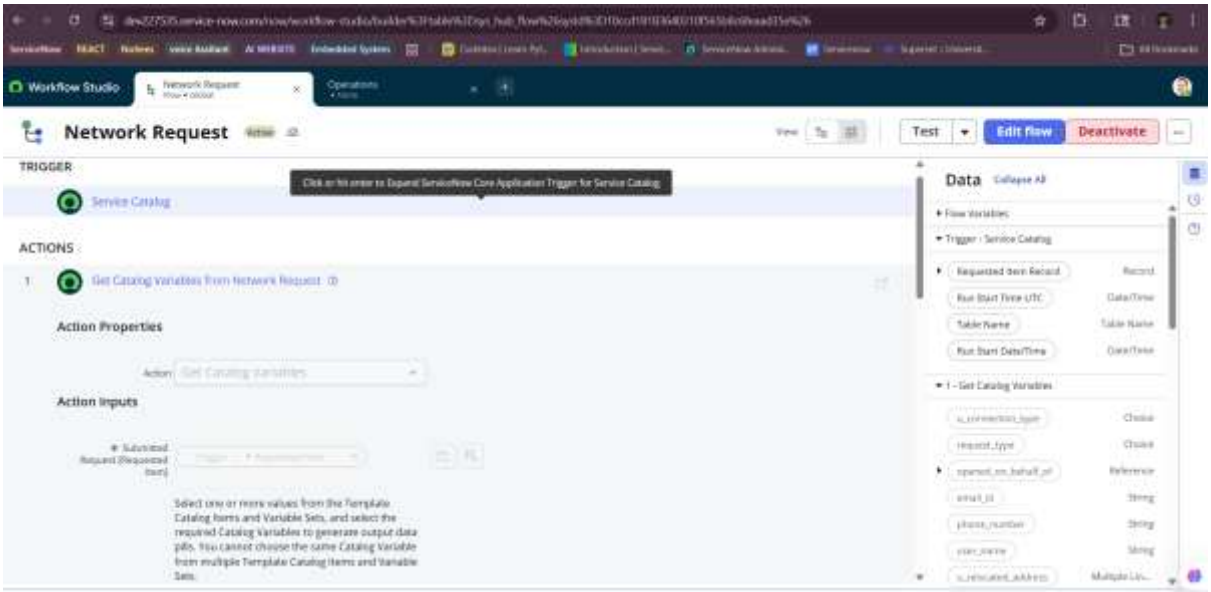
Data FLOW Designer Schema

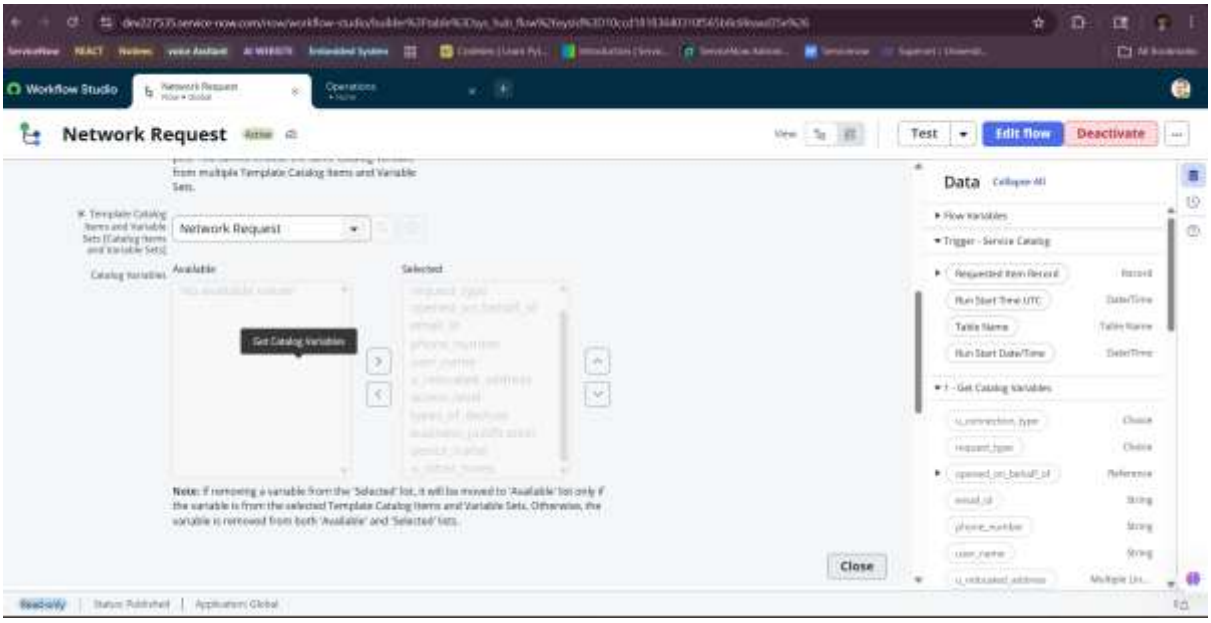
Catalog Variable	Database Field (u_network_database)	Logic / Automation
Opened on behalf of	u_requested_for	Auto-populated via Client Script
Device Details	u_device_details	Mandatory Field Enforcement
Status (N/A in portal)	u_work_status	Controlled by Flow Designer (Protected by ACL)

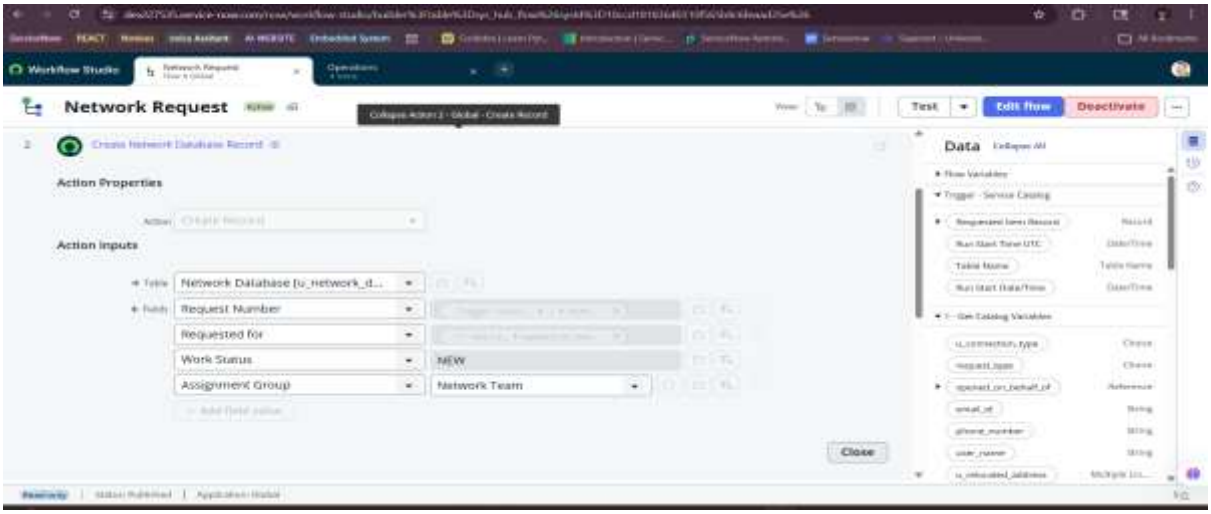


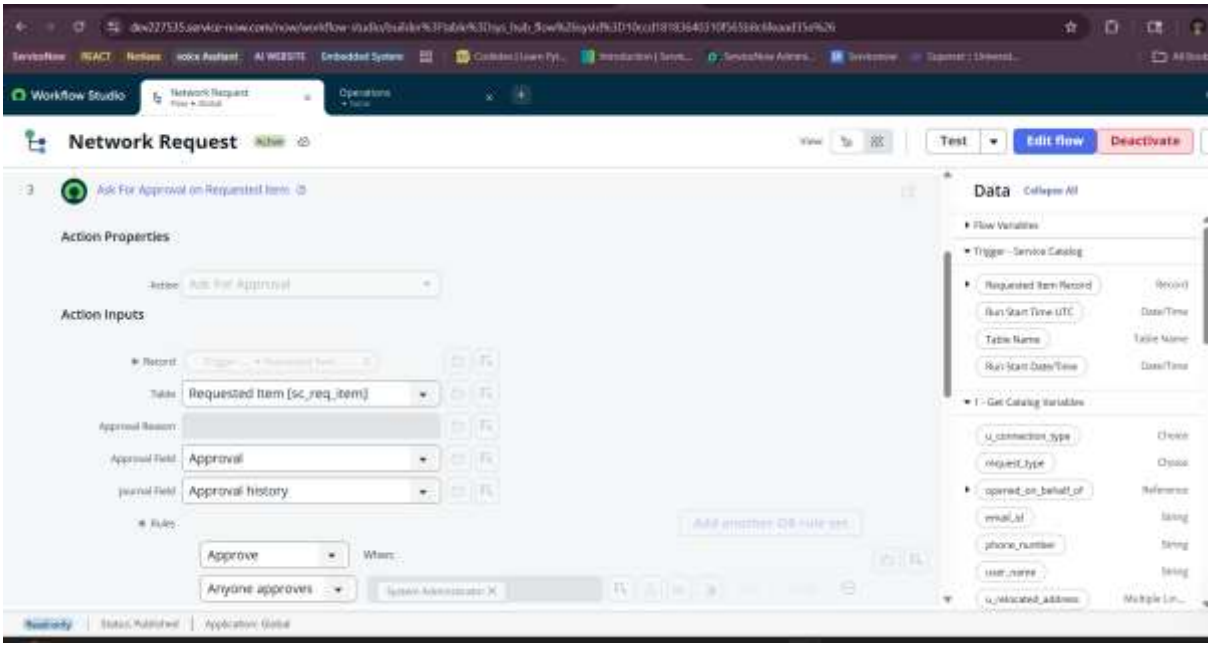


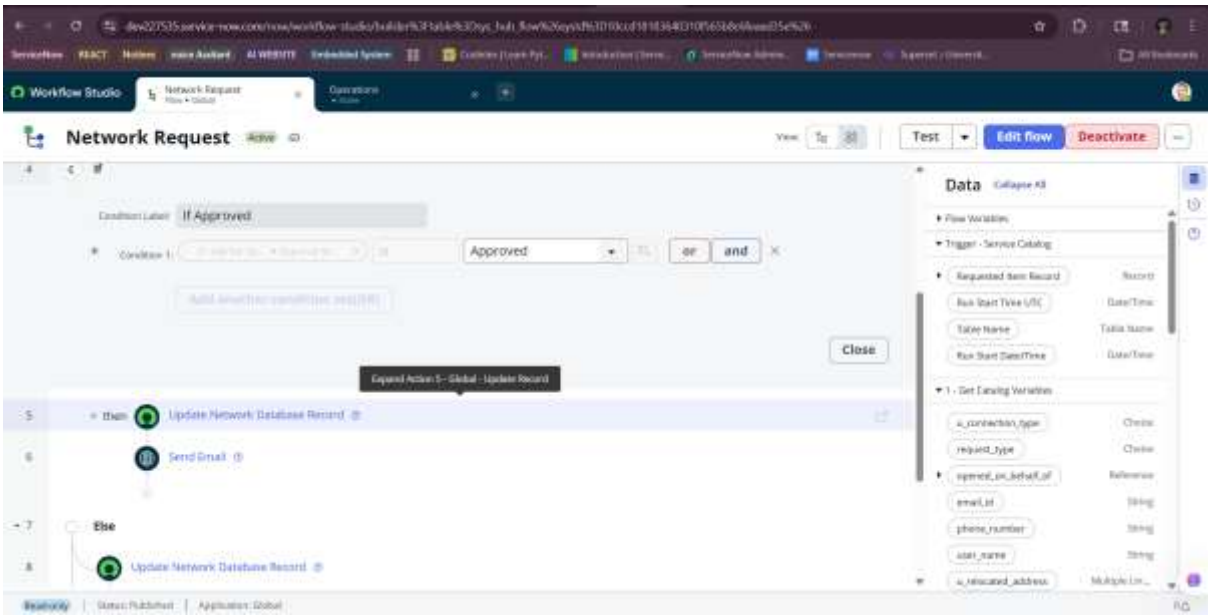

Workflow Studio
Flow executions
Export
Name
State
Runtime
Created by
Created
Sys ID
Network Request 320 system 2026-04-28 10:47:29 3d155edd5e200310af002c46f8c629f5
Network Request 1,431 system 2026-04-28 10:31:54 5d819ad548200310c1f0bad6ffdcb589
Network Request 992 admin 2026-04-28 11:19:18 bb5cd2dd46003107 8c921178563f958
Network Request 627 system 2026-04-27 12:55:54 bce821c925a403105de0e71c2f46a5d9
Network Request 1,789 admin 2026-04-28 11:18:26 d72cda9d736003103edcafe47673a1bb
Showing 1-5 of 5
rows per page

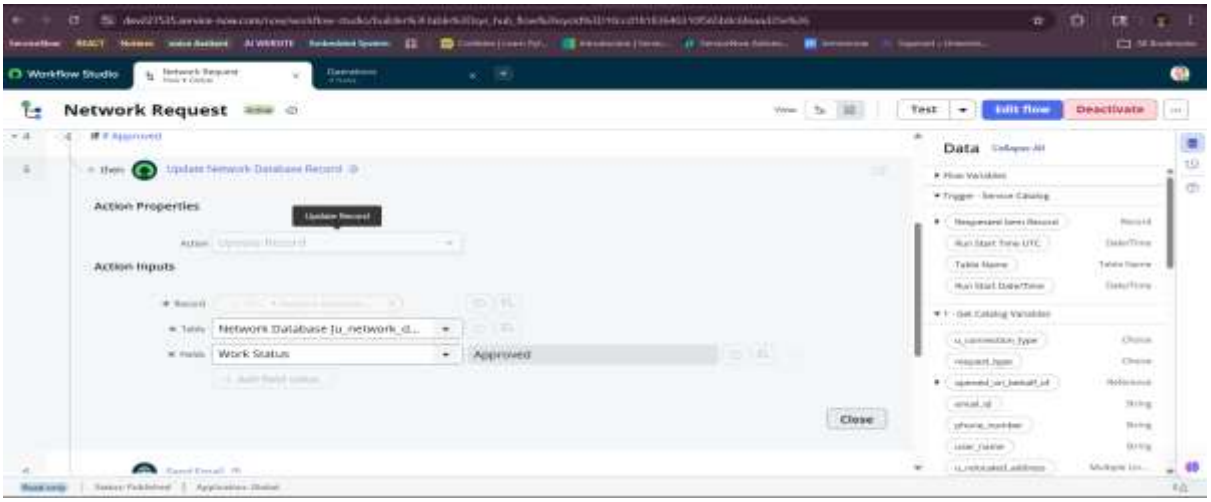

Workflow Studio
Network Request
Test
Edit flow
Deactivate
TRIGGER
Service Catalog
ACTIONS
Get Catalog Variables from Network Request
Action Properties
Action Inputs
Data

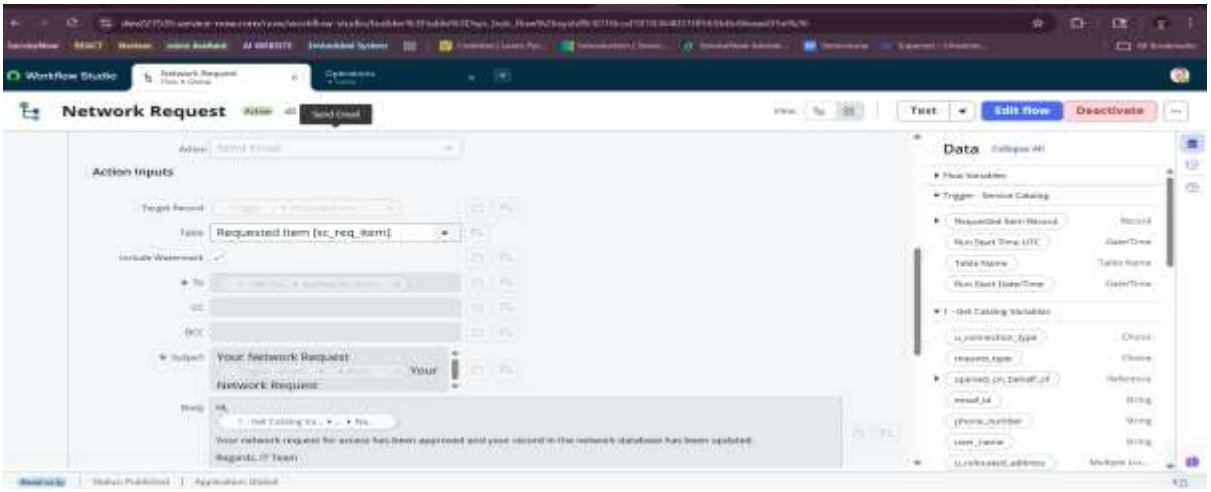

Workflow Studio
Network Request
Test
Edit flow
Deactivate
Network Request
Available
Selected
Get Catalog Variables
Data
Close

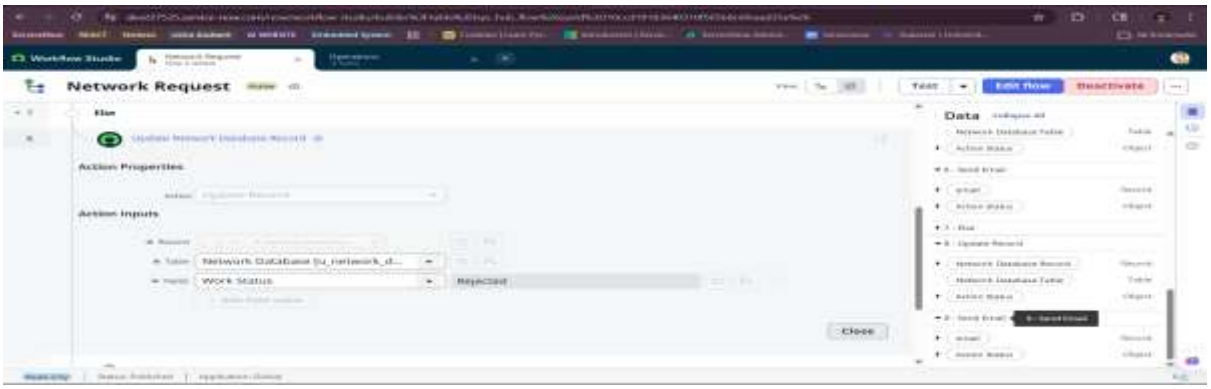

Workflow Studio
Network Request
Test
Edit flow
Deactivate
Collapse Action 2 - Global - Create Record
Create Network Database Record
Action Properties
Action Inputs
Network Database (u_network_d...
Request Number
Requested for
Work Status
NEW
Assignment Group
Network Team
Close
Data
Collapse All

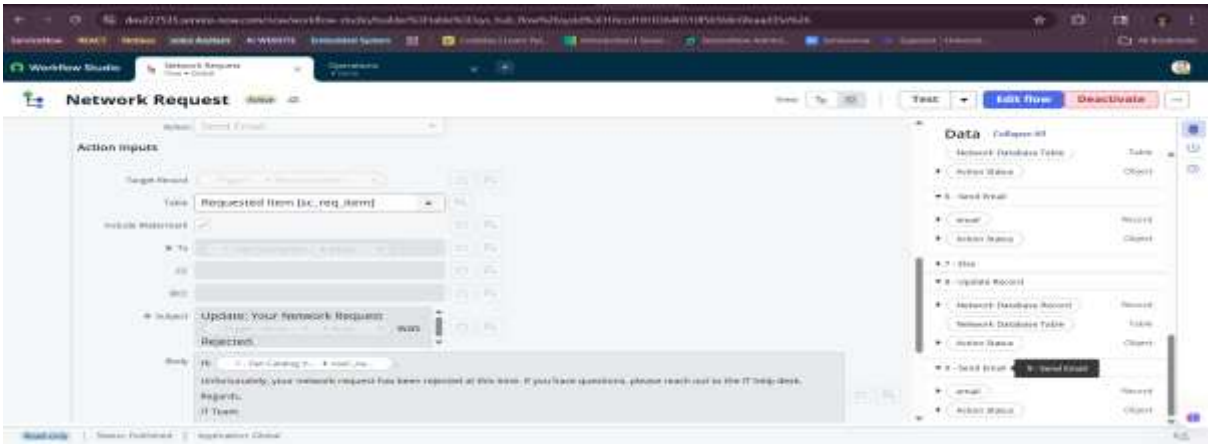

Workflow Studio
Network Request
Operations
Test
Edit flow
Deactivate
Ask For Approval on Requested Item
Action Properties
Action Inputs
Requested Item (sc_req_item)
Approval
Approval history
Approve
Anyone approves
Data
Collapse All


Workflow Studio
Network Request
Test
Edit flow
Deactivate
If Approved
Approved
or
and
Close
Expand Action 5 - Global - Update Record
Update Network Database Record
Send Email
Else
Update Network Database Record
Data
Collapse All


Workflow Studio
Network Request
Test
Edit flow
Deactivate
Then
Update Network Database Record
Action Properties
Update Record
Action Inputs
Network Database [u_network_d...
Work Status
Approved
Close
Data


Workflow Studio
Network Request
Send Email
Test
Edit flow
Deactivate
Action Inputs
Requested Item [sc_req_item]
Your Network Request
Your Network Request
Data


Workflow Studio
Network Request
Test
Edit flow
Deactivate
Else
Action Properties
Action Inputs
Network Database [u_network_d...
Work Status
Rejected
Close
Data


Workflow Studio
Network Request
Test
Edit flow
Deactivate
Action Inputs
Requested Item [sc_req_item]
Update: Your Network Request
Rejected
Data

3.Setup Manual (Recreation Guide)

Developer instructions for replicating this application in a new environment.

Step 1: Create Custom Table

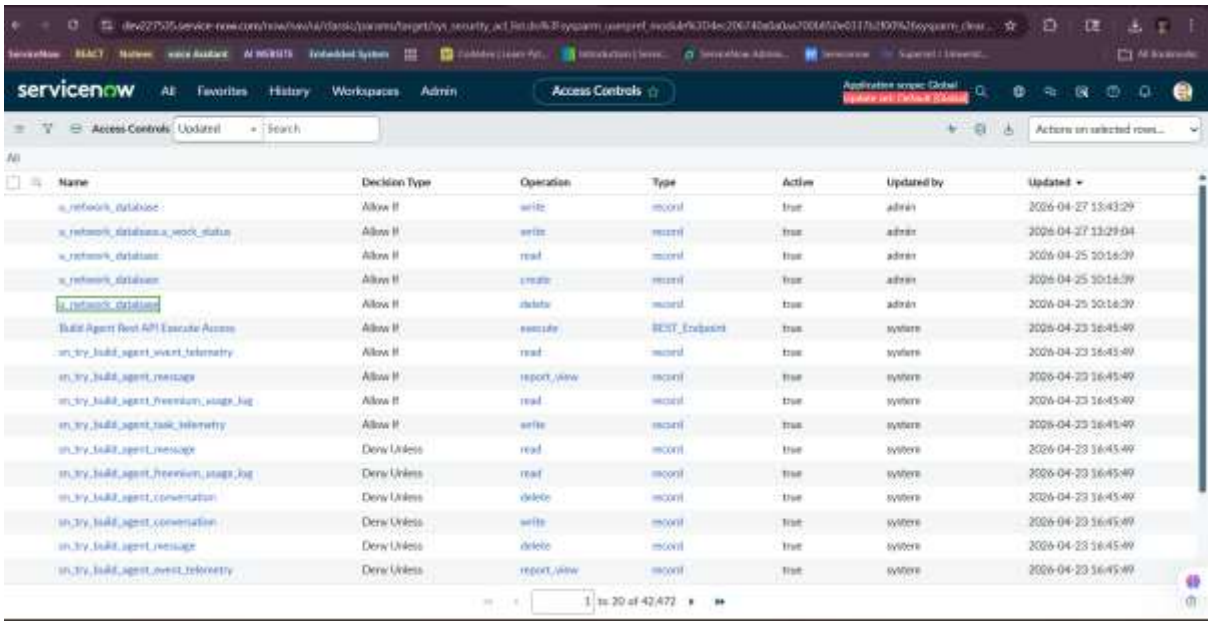
Navigate to **System Definition > Tables**. Name it *Network Database*. Add fields for Request Number (String) and Work Status (Choice).

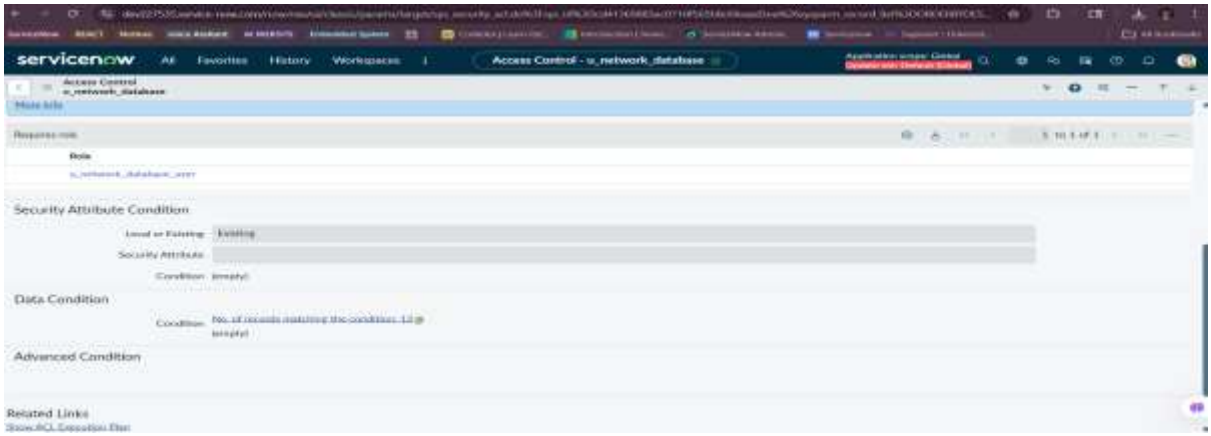
Step 2: Security Configuration (ACL)

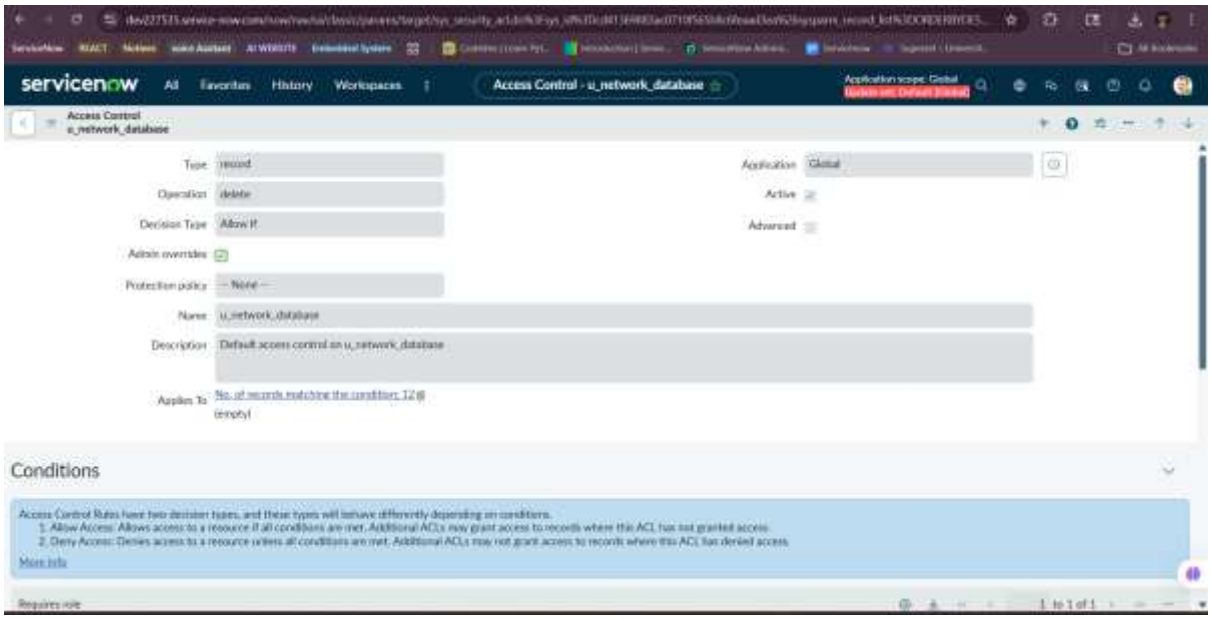
Create a **Write ACL** on the *Work Status* field. Restrict access to the `admin` role and **uncheck** "Admin Overrides" if you want it locked for everyone during testing.

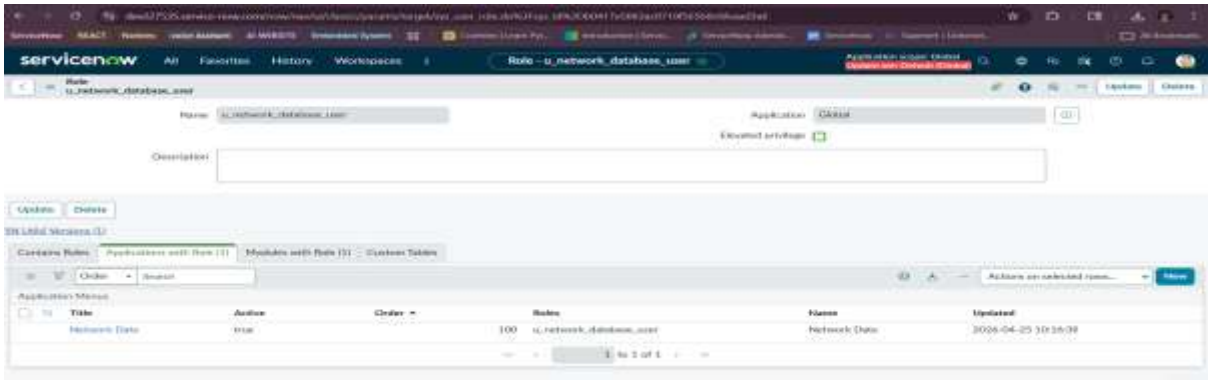
Step 3: Workflow Automation

Build the Flow in **Workflow Studio**. Ensure the *Ask for Approval* action is set to dotwalk to the *Assignment Group.Manager*.





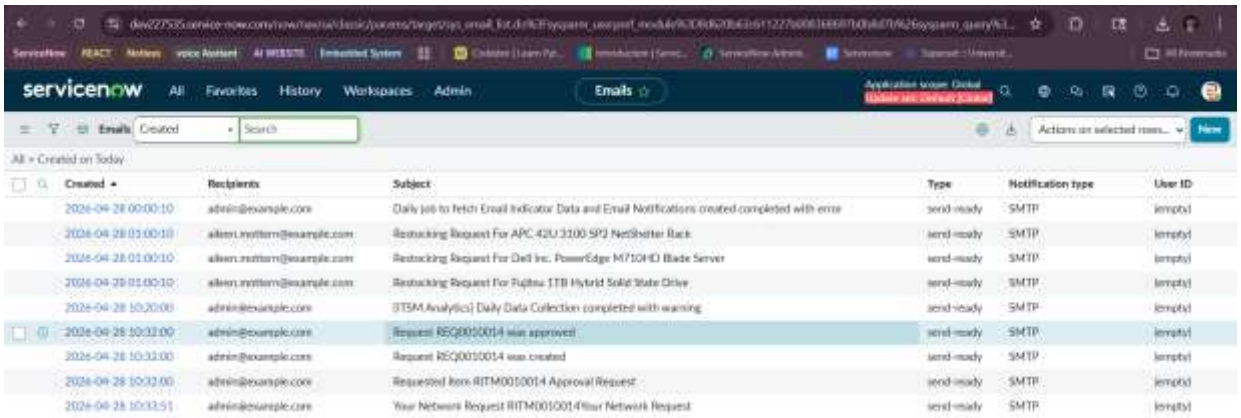


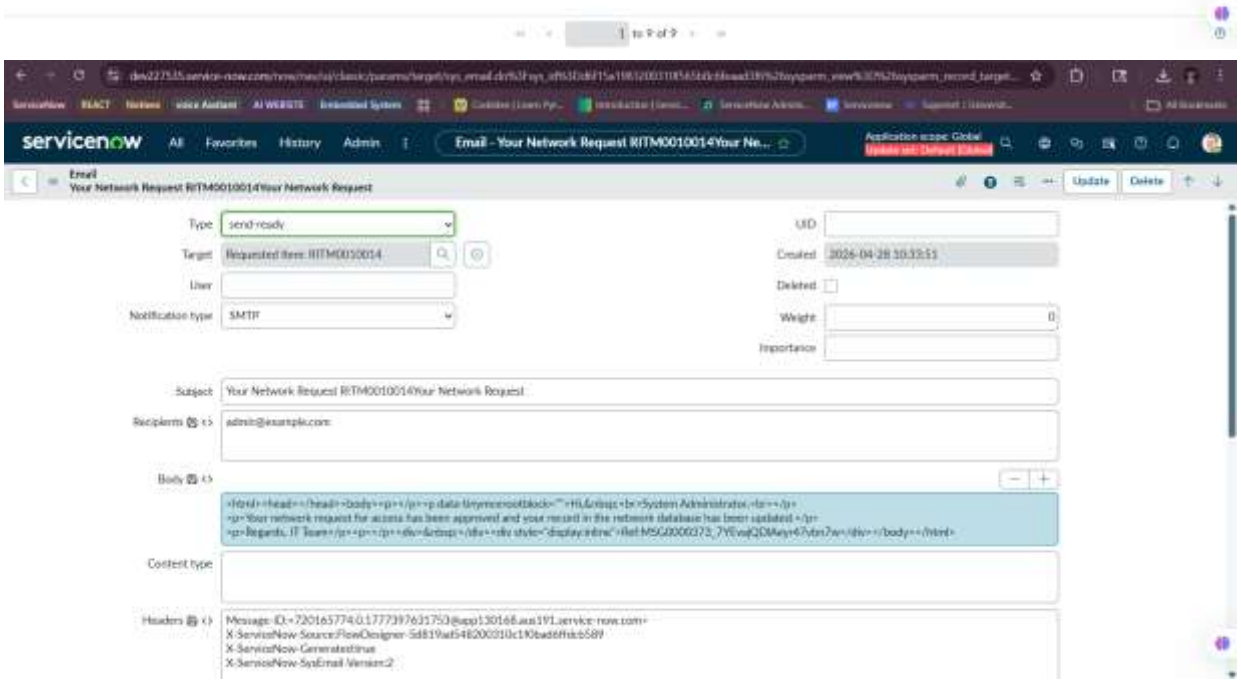


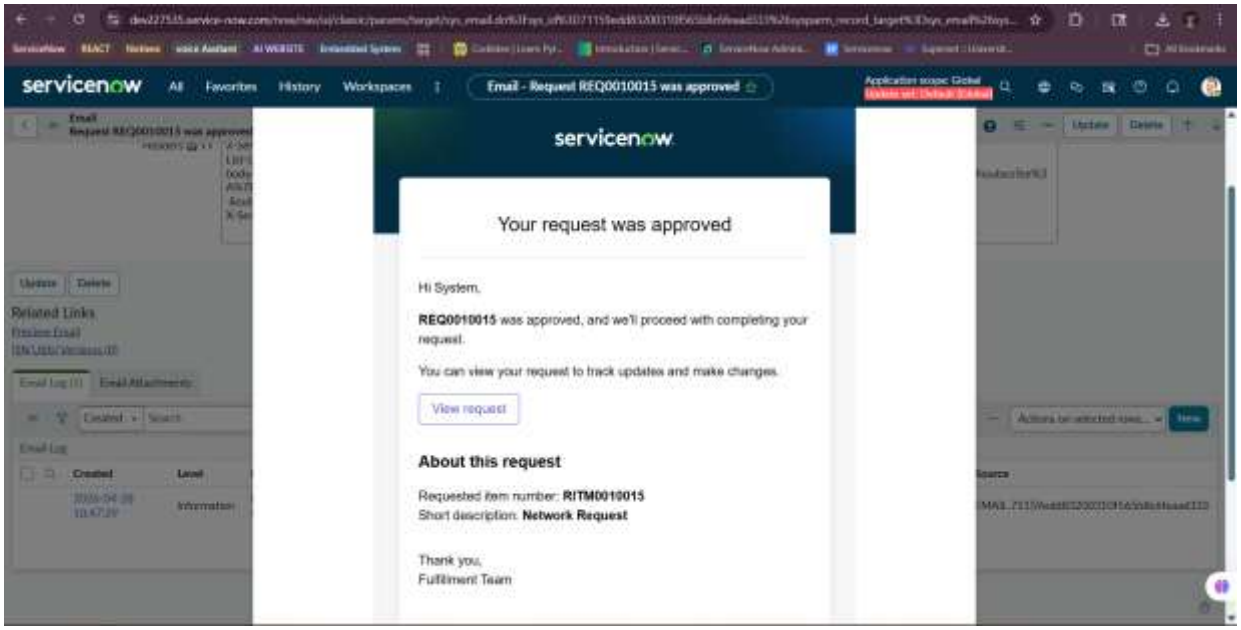
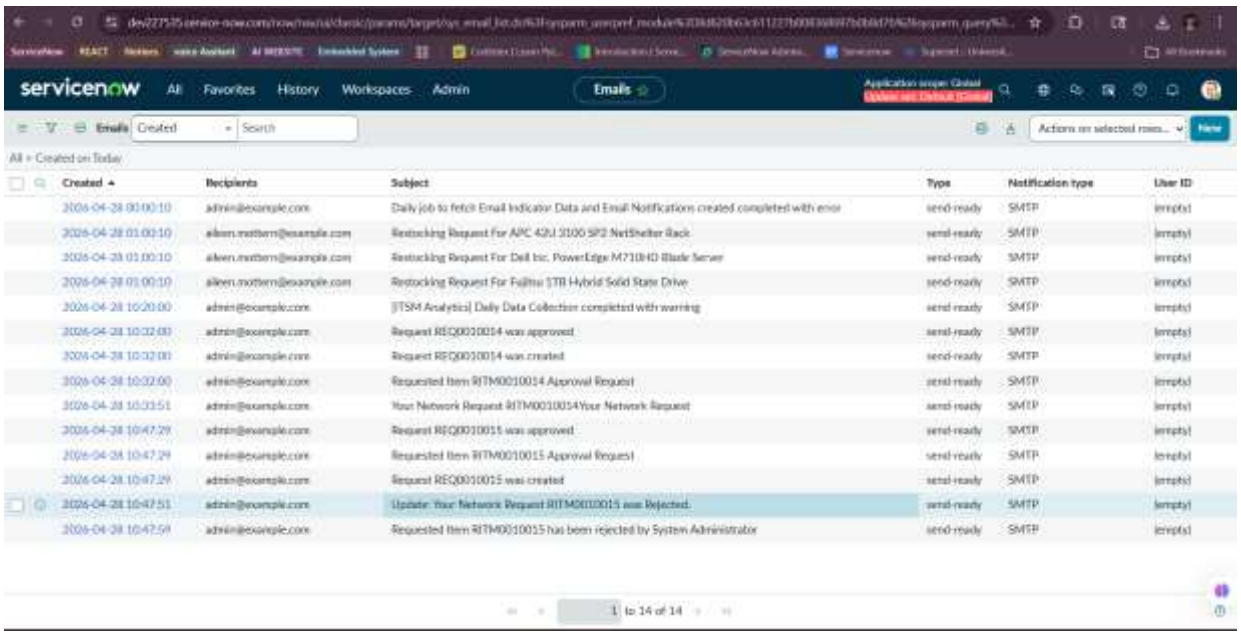
4.QA Testing & Validation

Verified the request lifecycle through rigorous end-to-end testing.

- **Happy Path:** Submitted request RITM0010013 was approved. Table status updated to 'Approved' and email sent.
- **Rejection Path:** Manager rejected request; record status correctly flipped to 'Rejected'.
- **Access Control:** Standard users received "Security Constraints" when attempting to bypass the workflow.



- 

- 
servicenow
All
Favorites
History
Workspaces
Email - Request REQ0010015 was approved
servicenow
Your request was approved
Hi System,
REQ0010015 was approved, and we'll proceed with completing your request.
You can view your request to track updates and make changes.
View request
About this request
Requested item number: RITM0010015
Short description: Network Request
Thank you,
Fulfilment Team
- 
servicenow
All
Favorites
History
Workspaces
Admin
Emails
Created
Recipients
Subject
Type
Notification type
User ID
1 to 14 of 14
-
servicenow
Preview Email